Extended essay: commentary for example essay			
DP subject: (Subject-focused essay)	**Mathematics**		
DP subjects: (Interdisciplinary essay)		Interdisciplinary framework:	
Topic:	**Calculating π**		
Research question:	**Which is the most efficient method of calculating π ?**		
Assessment details			
Criterion:	Mark awarded:	Commentary: (Incorporating wording from the assessment criteria)	
A: Framework for the essay [Maximum possible mark: 6]	**4**	Research questions (RQ) is partially focused. Appropriate research methods are used, but it is not clear why a seemingly random selection of 5 methods of calculating π is relevant. Why not just 2, explained more thoroughly? A suitable structure makes the essay well communicated.	
B: Knowledge and understanding [Maximum possible mark: 6]	**4**	Relevant research materials are used, but limits (p7) and Binomial Theorem (p19) are not well explained. Recursive formula, product sign, Squeeze Theorem are not explained (p11-13) suggesting this is copied from a source, rather than explained in the student's own words. Appropriate terminology and concepts are used, but understanding is not always shown.	
C: Analysis and line of argument [Maximum possible mark: 6]	**5**	A consistent line of argument with the RQ always in focus. Findings and conclusions are clear.	
D: Discussion and evaluation [Maximum possible mark: 8]	**5**	Limited discussion of the different methods supported by appropriate evidence, but candidate does clearly answer the RQ. Some discussion of limitations.	
E: Reflection [Maximum possible mark: 4]	**2**	Reflections are mainly descriptive, and largely confined to discovering Excel. Little on the mathematical methods used. Some evidence of the student's growth through the process.	
Total marks awarded	20/30		
Holistic comment on the essay:	A well written essay, some good research carried out, but candidate would have been better advised to look at fewer methods, and to explain the mathematics behind them more thoroughly.		